

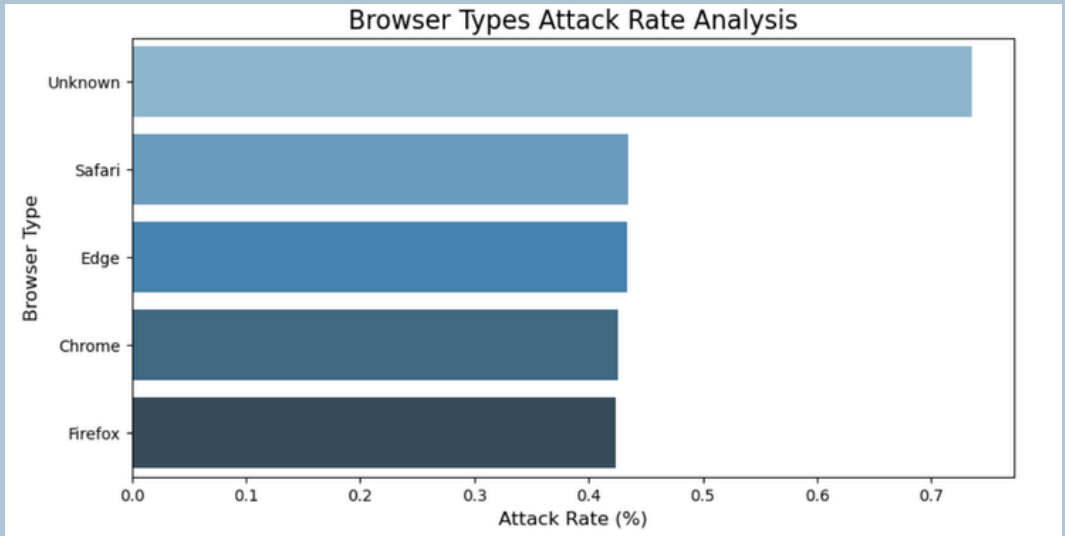


Cybersecurity Intrusion Data Set

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Introdcution: Dataset represents individual network sessions. Each row is a single session with information like login attempts, encryption, browser type, and whether an attack was detected.

Independent: Browser Type **Dependent:** Attack Detected



Which browser types are most frequently targeted in cyber attacks?

The attack rate is highest for "Unknown"(could be an indicator of automated scripts or bots) browser types.

Model's predictions have an average error of 18%, and it explains only 26% of the factors leading to attacks.
Conclusion: model's performance is limited & requires improvement.

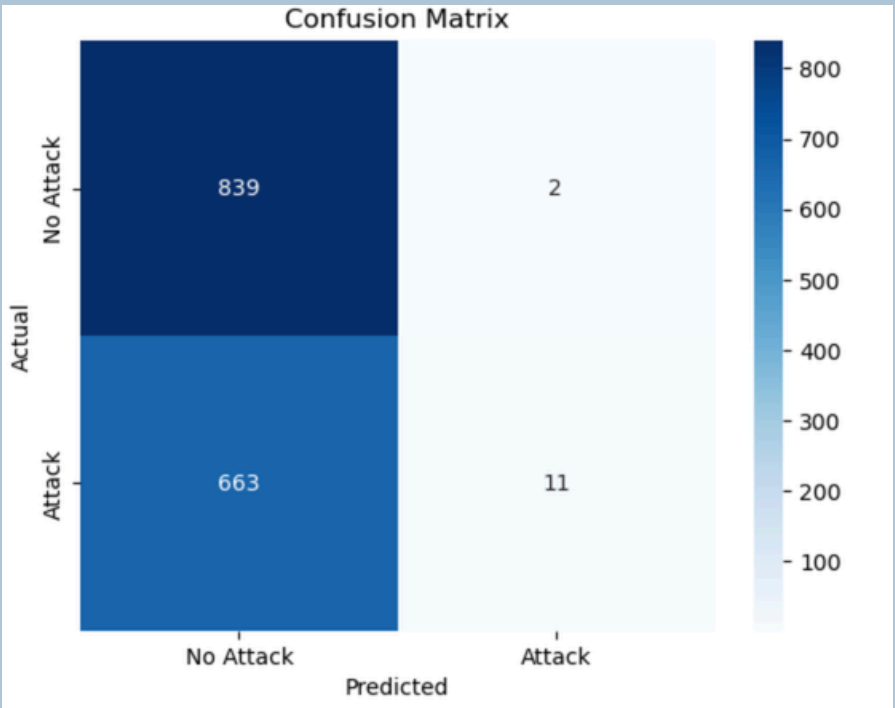
Mean Squared Error: 0.18

R² Score: 0.26

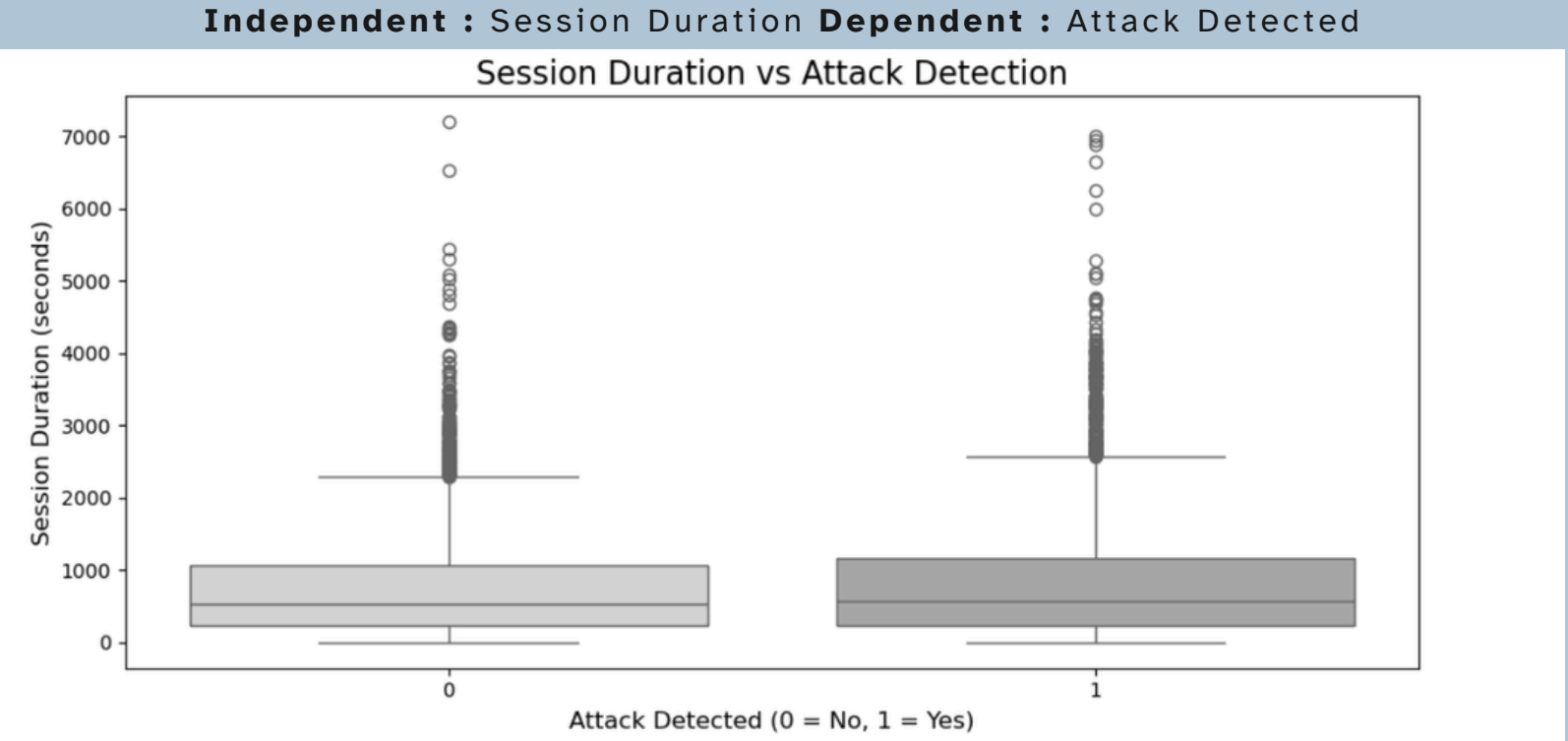
Accuracy: 0.5789

Confusion Matrix:
[[818 23]
[615 59]]

How do session durations relate to the likelihood of a cyberattack?
Hypothesis: Longer session durations are more likely associated with cyberattacks.



Confusion Matrix:
True Negatives: 839 // False Positives: 2 // False Negatives: 663 // True Positives: 11



R² (-0.777): model explains -77.7% of the variance(performs worse than random guessing)
MSE (0.438): model's predictions are off by 0.438, indicating substantial prediction error.

Conclusion: model is underperforming, struggles to detect attacks (with many false negatives), and session duration alone is not a strong predictor. incorporating more relevant features (IP reputation, protocol type), model's ability to accurately detect cyberattacks could significantly improve
Future consideration: dataset doesn't show strong correlations between key features (including encryption used, failed logins, and attack detection), which limits the model's ability to predict attacks accurately. adding more relevant features could help improve future performance.

